RPA Dashboard: Calculations and Options Documentation

This document outlines the calculation logic for key metrics and the options available to users within the RPA Dashboard application.

1. Calculations & Metrics

The application aggregates data from individual Bot runs and static configurations to present real-time dashboards.

1.1. Organization Level Metrics

Reflects the overall state of the automation program (visible on Landing Page & Admin Dashboard).

Metric	Calculation Logic
Total Bots	Count of all bots in the database, regardless of status.
Deployed Bots	Count of bots where status is one of: Deployed, deployed, Live, live.
Total Departments	Unique count of departments associated with bots.
Total SPOCs	Unique count of Single Points of Contact (SPOCs).

Man Hours Saved

- **Formula**: Total Hours Saved / 8
- *Explanation*: We assume a standard 8-hour workday. Every 8 hours of bot time saved equals 1 Man-Day.

Total Hours Saved (Hybrid Logic)

The dashboard uses two different methods to calculate total savings based on the bot's schedule type:

1. **On Demand / Multiple Schedule Bots**:
    - **Logic**: Run-Based. Savings are only added when the bot *actually runs successfully*.
    - **Formula**: Total Hours Saved = (Number of Successful Runs) * (Value Per Run)

    - *Note: If the bot does not run, no hours are added.*
2. **Scheduled Bots (Daily/Weekly/Monthly)**:
    - **Logic**: Time-Based Projection. Savings accumulate automatically over time based on the deployment date, regardless of individual run logs (to preserve historical data).
    - **Formula**: Total Hours Saved = (Days Since Deployment) * (Monthly Hours / 30)

Hours Saved Today

- **Formula**:
- **(Successful Runs Today) * (Value Per Run)**
- *Note: This metric is always based on actual run data from the daily report.*

Daily Statistics (Today)

- **Hours Saved Today**: Successful Runs Today * Value Per Run.
- **Run Status**:
    - **Success**: At least one run today was "Completed".
    - **Failed**: At least one run today "Failed" (and none "Completed").
    - **Not Run**: No runs recorded for today.

1.3. Department Level Metrics

Aggregates data for a specific department.

- **Active Bots**: Count of bots in the department with status Deployed, Live, or Active.

- **Inactive Bots**: Total Bots - Active Bots.
- **Total Hours Saved**: Sum of total_hours_saved (calculated column) for all bots in the department.
- **Man Hours Saved**: Total Hours Saved / 8.
- **Growth Metrics**: Percentage change in Hours/Man Hours compared to ~30 days ago (based on chart history).

2. User Options & Features

2.1. Landing Page

The entry point for all users.

- **Global Search**: (Implicit via department selection)
- **Sync Data**:
    - **Sync Now Button**: Triggers a manual synchronization with SharePoint/Excel sources to fetch the latest bot run logs. Includes a status indicator (Success/ Failed).
- **Navigation**:
    - **Department Cards**: Click to drill down into specific department dashboards.
    - **Interactive Org Chart**: "View Interactive Org Chart" button for a hierarchical view.
    - **Admin Console**: Lock icon (top-right) to access backend management.
- **Department Sorting/Cards**: Displays key metrics (Active Bots, Hours Saved) per department.

2.2. Department Dashboard

Detailed view for a selected department.

- **Filters**:
    - **Search Bar**: Key text search across Bot Name and Use Case Name.
    - **Status Filter**: Dropdown to filter list by:
        - **All Status**
        - **Active** (Deployed/Live)
        - **Inactive**
- **Charts**:
    - **Hours Saved Variation**: Area chart showing cumulative hours saved over time (monthly trend).
    - **Man Hours Saved Variation**: Area chart showing cumulative man hours.
    - **Info Modals**: Click "Info" icon on charts to view calculation explanations (Odometer vs Speedometer analogy).
- **Bot List**:
    - **Sort/View**: Table listing all bots.
    - **PDD Link**: Direct link to Process Definition Document if available.
    - **View Action**: Opens a detailed modal with run history, schedules, and specific logic for that bot.

2.3. Admin Console

Restricted area for managing the system.

- **Dashboard Tab**:
    - System Health (DB Connection, SharePoint Status).
    - Recent Sync Activity log.
- **Bot Management Tab**:
    - **Add New Bot**: Form to manually register a new bot.
    - **Edit Bot**: Modify existing configurations (Hours, Schedule, etc.).
    - **Delete Bot**: Remove a bot from the system.
    - **Filters**: Search by name/department; Filter by specific Status.
- **File Logs Tab**:
    - Audit trail of all processed files.
    - **Download**: Ability to download the original processed files for verification.
- **Configuration Fields (Bot Form)**:
    - Agent Code & Name
    - Department & SPOC assignment
    - Status (Active/Deployed/Hold/etc.)
    - **Calculation configs**: Monthly Hours, Per Day Saving, Frequency.
    - Metadata: PDD Link, Developer Name, Schedule Details.

3. Data Dictionary

Term	Definition
SPOC	Single Point of Contact – The person responsible for a bot/process.
PDD	Process Definition Document – Technical/Functional spec of the bot.
Man Hour	Equivalent of one person working for one hour. Standardized as 1/8th of a saved bot hour.
Man Month	(Reference) Equivalent of one person working for a month. calculated as Total Hours / 240 (8 hours * 30 days).

4. Data Ingestion & Smart Matching Logic

The system includes robust logic to ingest data from Excel files and match inconsistent naming conventions between the Master Bot List and Daily Run Reports.

4.1. Bot Identification Strategy

When processing Daily Run Reports, the system attempts to match the "Activity Name" to a known Bot using the following waterfall approach:

1. **Exact Match**: Direct string match with Use Case Name.
2. **Prefix Match**: Matches based on the text before the first underscore _ or pipe |.
    - *Example:* AGEL001_Invoice_Process *matches with* AGEL001.
3. **Fuzzy/Contains Match**: Checks if one name contains the other.

4.2. Value Normalization

- **Time Units**: Automatically parses "10 min", "2 hr", etc., into standard float Hours.
- **Dates**: Handles multiple date formats (DD-MM-YYYY, MM/DD/YYYY, etc.) to ensure consistent timeline tracking.